

BQ79616-Q1 Daisy Chain Communications Application Report



ABSTRACT

This application report is designed to demystify the bq7961X daisy chain communications interface when stacking multiple devices for high voltage applications. The goal is to provide a deeper understanding of the protocol details when evaluating the device. Additionally, this report provides recommended isolation components and typical application practices for system solutions to take full advantage of designing with this interface.

Table of Contents

1 Introduction	3
2 Receiver Topology and Common Mode Voltages	5
3 Signal Integrity Requirements	6
3.1 Receiver Timing Requirements	6
3.2 Receiver Threshold Requirements	6
4 Debug Fault Registers	8
5 Isolation Types	9
5.1 Transformer Isolation	9
5.2 Capacitor Only Isolation	9
5.3 Capacitor + Choke Isolation	10
6 Mixed Isolation Circuits using bq79600-Q1	11
7 Ring Architecture	13
8 Noise Immunity and Emissions	15
9 Daisy Chain Cable Selection	15
10 References	15

List of Figures

Figure 1-1. Daisy Chain Byte Level Definition	3
Figure 1-2. Communication Tones	4
Figure 1-3. Daisy Chain Bit Level Definition	4
Figure 2-1. Common Mode Voltage Diagram	5
Figure 3-1. Differential Signal Eye Diagram - 1.7V VSTART	6
Figure 3-2. Differential Signal Eye Diagram - 1.6V VSTART	7
Figure 3-3. Differential Signal Eye Diagram - BQ79600	8
Figure 5-1. Transformer Isolation Across PCBs	9
Figure 5-2. Capacitors Only Across PCBs	10
Figure 5-3. Capacitor Only on Same PCB	10
Figure 5-4. Capacitor + Choke Across PCBs	10
Figure 6-1. Example: BQ79600-Q1 Connected to BQ79616-Q1 Timing	11
Figure 6-2. How a Stack Command is Sent	11
Figure 6-3. When Does Each Device Receive the Command?	12
Figure 6-4. ??	12
Figure 7-1. Example to Change Communication Direction in Daisy Chain	13
Figure 7-2. Using Ring Architecture to Access All Devices in a Broken Cable Case	14

List of Tables

Table 3-1. Timing Chart	7
Table 5-1. Recommended Transformer Part Numbers	9

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1 Introduction

The daisy chain communication interface on the bq7961X family of devices is a proprietary protocol developed by Texas Instruments. It is designed using differential signaling to minimize Electro-Magnetic Susceptibility (EMS) and strengthen Bulk Current Injection (BCI) immunity. The differential communication transmits complement data on the COM*P and COM*N pins, respectively. This interface is bi-directional and half duplex, and, therefore, has a transmitter (TX) and receiver (RX) on the COMH (high -side) and COML (low-side) interfaces.

The device supports the use of transformers or capacitors to electrically isolate the signals between devices in the stack. There are configurations where the devices are physically located on the same board or located in entirely separate packs connected with twisted pair wiring. For applications that have multiple devices on the same PCB, a single capacitor is connected between the COMH/L pins of the devices. For extremely noisy environments and stringent EMI/EMC requirements, additional filtering may be necessary. For devices that are separated by cabling, additional isolation components are used.

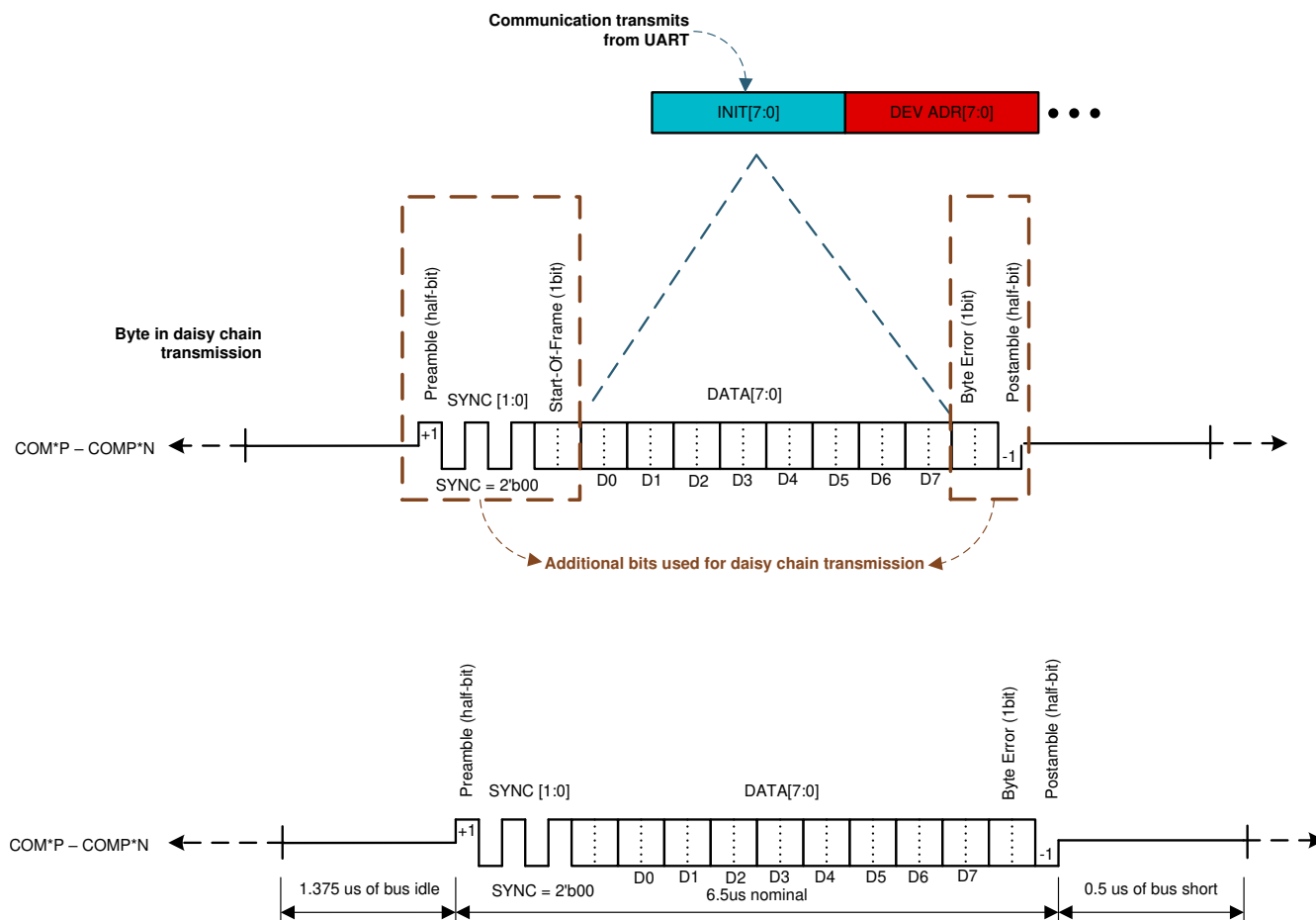


Figure 1-1. Daisy Chain Byte Level Definition

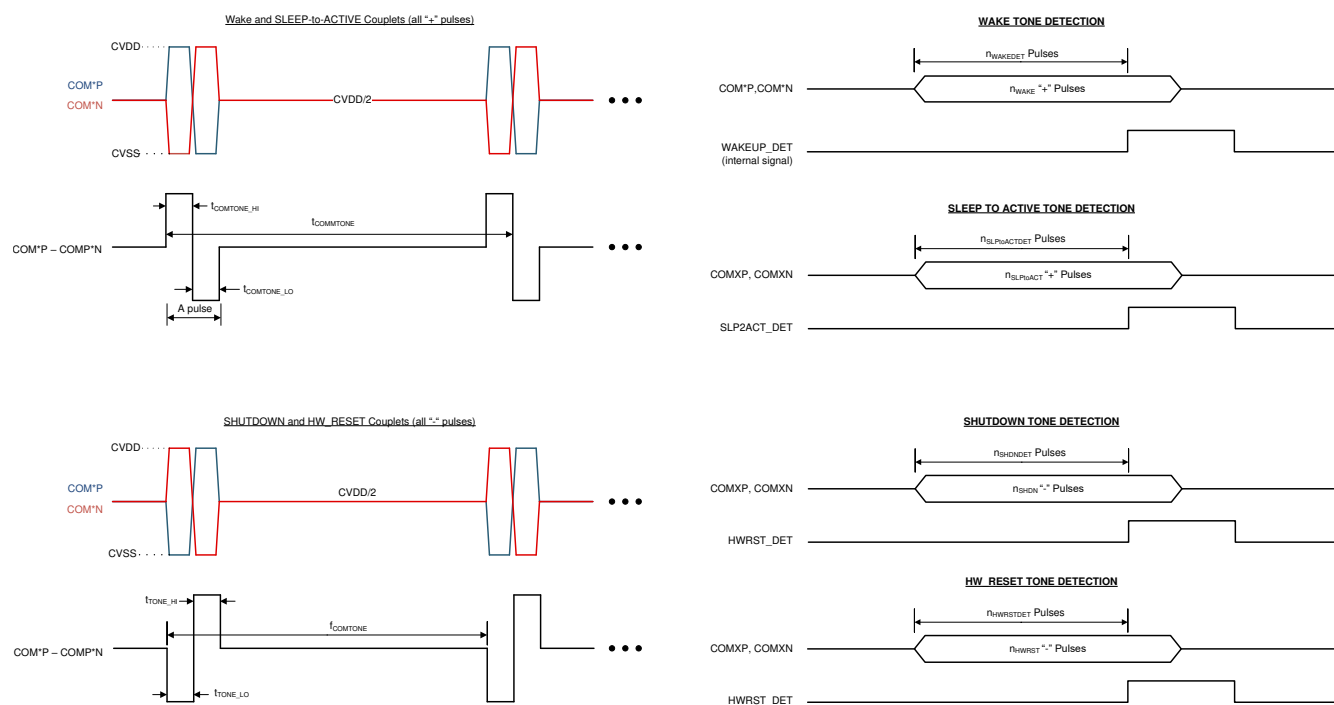


Figure 1-2. Communication Tones

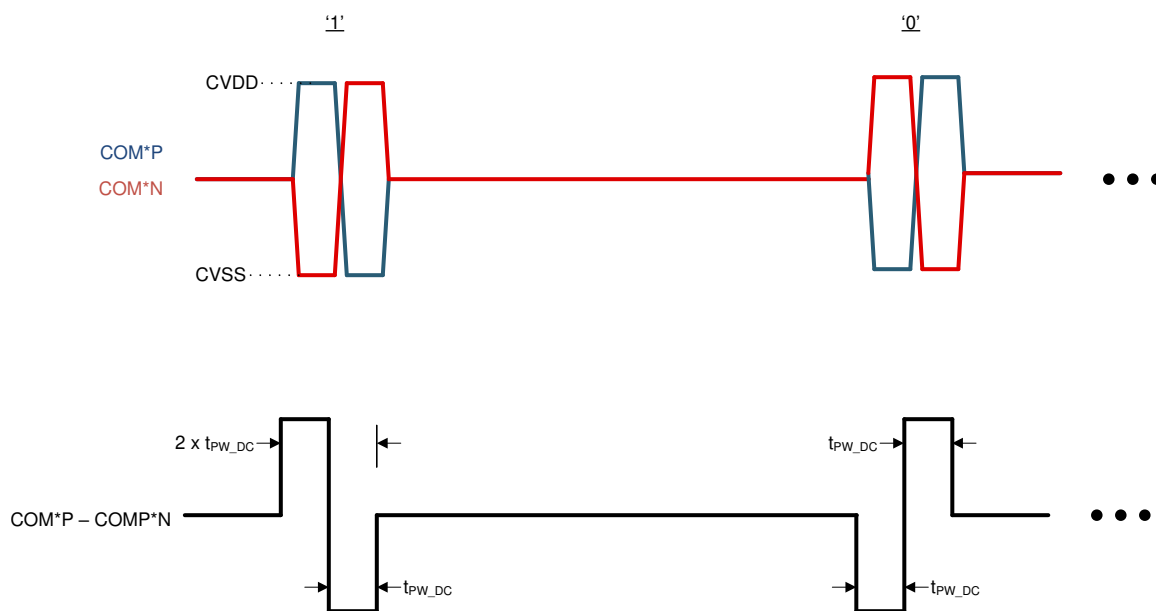


Figure 1-3. Daisy Chain Bit Level Definition

2 Receiver Topology and Common Mode Voltages

The daisy chain is bi-directional and half duplex, and, therefore, has a transmitter (TX) and receiver (RX) on the COMH and COML interfaces. The TX and RX functions are controlled automatically by the hardware based on the device's base/stack detection. When a WAKE ping/tone is received, the communication direction is set by CONTROL1[DIR_SEL] and the COMM_CTRL[TOP_STACK] configurations. See Start Communication in the bq79616-Q1 data sheet for details. Additionally, a user overwrite to take over the complete control of the COMH and COML is available under communication debug mode using the DEBUG_CTRL_UNLOCK, DEBUG_COMM_CTRL1, and DEBUG_COMM_CTRL2 registers. See the Debug Control and Status section for more details in the bq79616-Q1 data sheet. Additionally, for a more detailed explanation of the basic timings and function of the protocol and plots below, visit section 9.5.1.2. of the bq79616-Q1 data sheet.

The RX topology of the bq7961X devices is similar to RS485 but with added design mechanisms to attenuate common mode voltages of up to $\pm 20V$ due to noisy conditions. This common mode voltage limit for the differential signals is best explained by the chart in [Figure 2-1](#). In an ambient test scenario, the common mode voltage of both the COM*P and COM*N holds at 2.5V when there are no communications occurring. This is necessary because the attenuation of a negative signal would otherwise not bring the voltage between the receivers local ground and 5V CVDD rail (internal LDO dedicated for communications). If communications occur, then the device will drive the COM pins with the CVDD rail between 0V (low) and 5V (high) and return to 2.5V between data packets (idle time) and after the communications have completed.

In the event of a common mode voltage swing due to noisy conditions such as EV inverter noise or Bulk Current Injection (BCI) testing – the common mode voltage seen at the COM pins may oscillate up to $\pm 20V$ depending on the isolation components and current injected. The device is designed to withstand this type noise level up to $\pm 20V$ because of the internal attenuation that allows the 5V transceivers to handle the response and therefore the differential signal would still remain intact. Above this $\pm 20V$ level, there is risk of clamping the internal ESD structure or causing damage to the COM pins. This limit should be taken into consideration when determining BCI limits and testing.

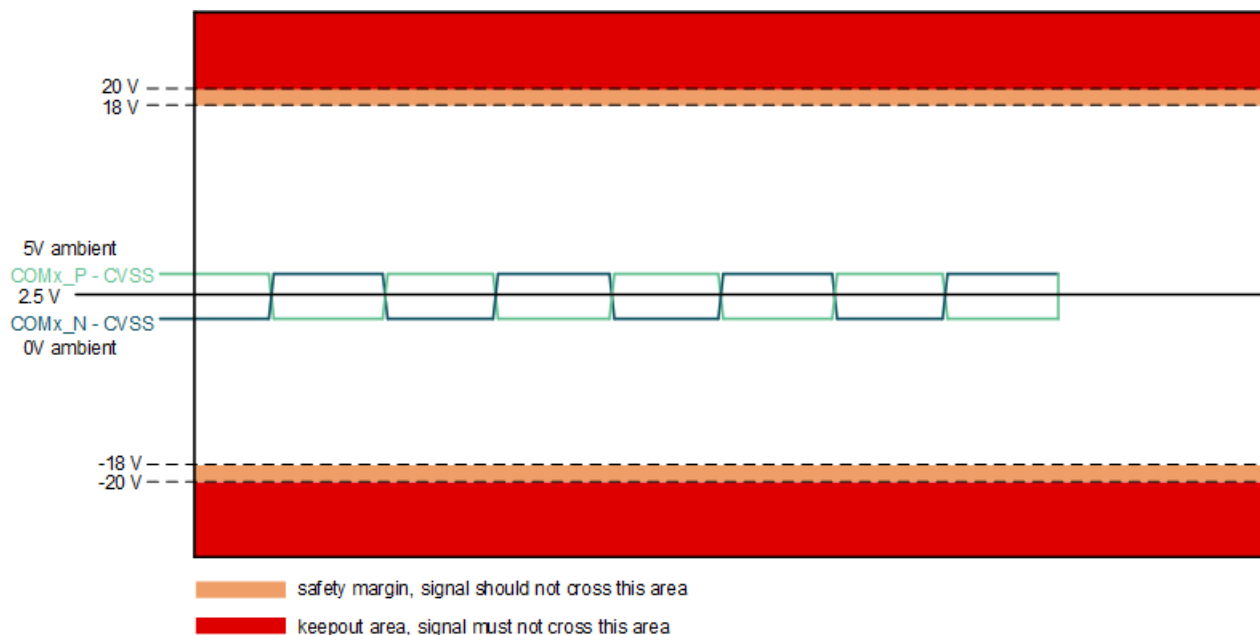


Figure 2-1. Common Mode Voltage Diagram

3 Signal Integrity Requirements

Proper receiver waveform timing and thresholds are critical factors for the internal digital to correctly interpret tones and data communications. Note that these factors may be affected by isolation component choices and cabling.

3.1 Receiver Timing Requirements

As shown below, the high and low timing differs between a tone signal (1 μ s pulsewidth at 500kHz) or data communications (~250ns pulsewidth at 2MHz) waveform. However, both tones and data communications share similar rise/fall times and must both be rectangular waveforms with a maximum decay below 1V. Note that the timing for the $t_{\text{high}}/t_{\text{low}}$ refers to the time that the signal must be above or below the proper threshold value for the integration to succeed. If the signal is noisy during this $t_{\text{high}}/t_{\text{low}}$ time and droops below the threshold value for some time - then there is risk of the signal being detected. We expect that roughly 1 μ s $\pm$ 8% for tones and 250ns $\pm$ 20ns for data communications is the allowed timing variation for these signals to still be properly detected.

3.2 Receiver Threshold Requirements

The thresholds at which the internal digital is able to properly detect a high vs. low transition are shown in . A good quality waveform will have a differential voltage above 1.8V (and below -1.8V on the negative side). It typically resembles a square wave with rounded edges.

The image below shows a poor quality or drooping waveform that still meets the criteria for a valid communication. The first diagram is showing a starting voltage (VSTART) of 1.7V with an ending voltage (VEND) of 1.2V. The second diagram shows a second scenario where the starting voltage (VSTART) is only 1.6V with an ending voltage (VEND) of 1.3V. If the starting amplitude is not at least 1.6V, then the bit will not be detected, regardless of ending amplitude. These types of drooping waveforms most often are seen when there are two adjacent bits of opposite polarity, forming what can be referred to as a double-bit.

Note: The color red below indicates the region that the signal may or may not be detected (poor quality waveform zone) whereas orange indicating the area the area near the triggering threshold range. A good quality waveform would remain in the green region above 1.7V for the duration of the positive pulse width.

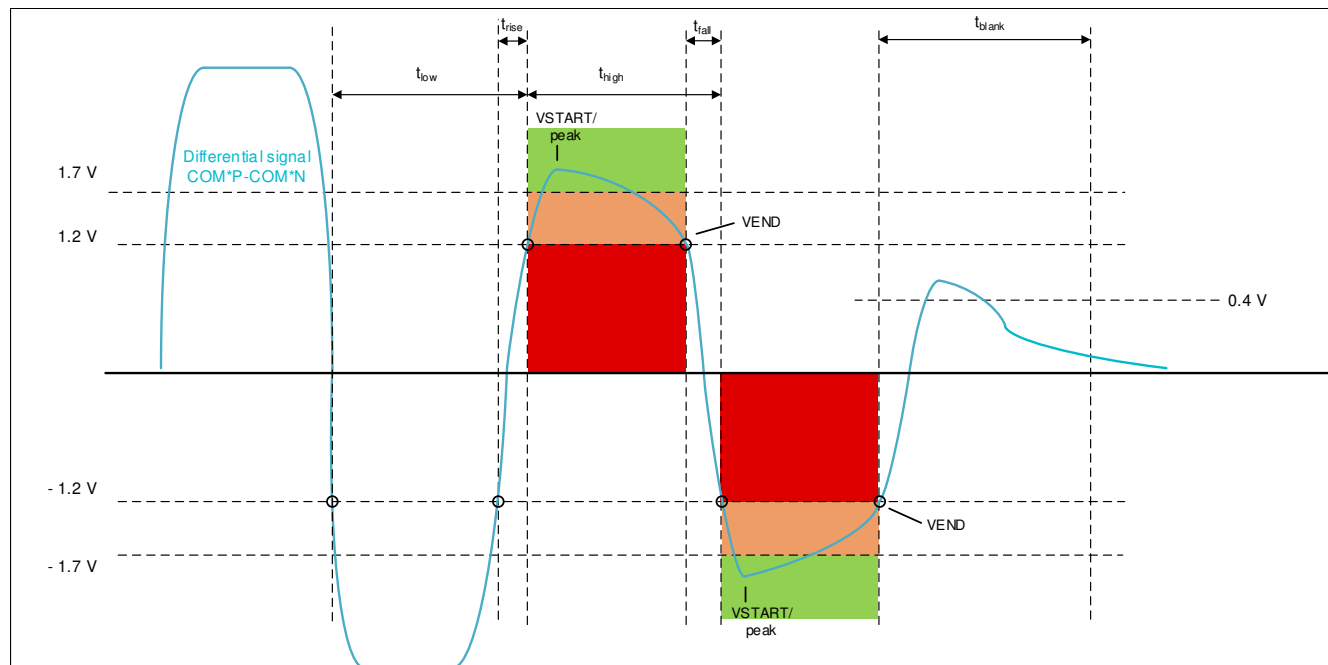


Figure 3-1. Differential Signal Eye Diagram - 1.7V VSTART

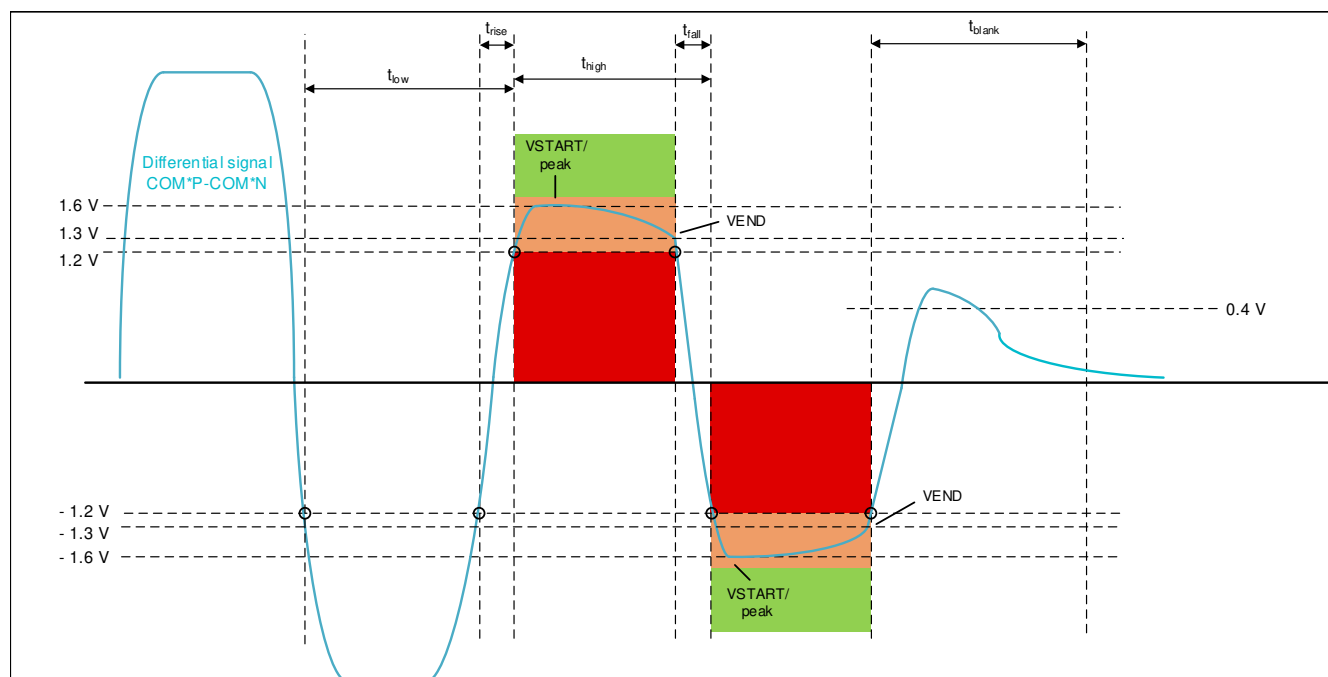


Figure 3-2. Differential Signal Eye Diagram - 1.6V VSTART

The table below gives the specific timing and voltage requirements for valid daisy-chain communication.

Table 3-1. Timing Chart

	Parameter	Conditions	Min	Typ	Max
t_{high}	positive pulse width	COMP-COMN from +1.2V of rising edge to -1.2V of next falling edge	220 ns	250 ns	280 ns
t_{low}	negative pulse width	COMP-COMN from -1.2V of rising edge to +1.2V of next falling edge	220 ns	250 ns	280 ns
t_{rise}	Rising slew rate	COMP-COMN from -1.2V to +1.2V of rising edge			60 ns
t_{fall}	Falling slew rate	COMP-COMN from +1.2V to -1.2V of falling edge			60 ns
V_{start}	Starting amplitude	If ending amplitude is above 1.3V	1.6 V		
V_{start}	Starting amplitude	If ending amplitude is between 1.2V and 1.3V	1.7 V		
t_{blank}	Blanking time	After last bit	4 μ s		
Wake Tone (t_{high}/t_{low})	Wake Tone pulse widths	COMP-COMN from +1.2V of rising edge to -1.2V of next falling edge during and vice versa for Wake tones	1 μ s		

This last diagram shows the threshold needed for interfacing with the BQ79600 device. The primary difference is the threshold and timing references are based on a 1.8V threshold.

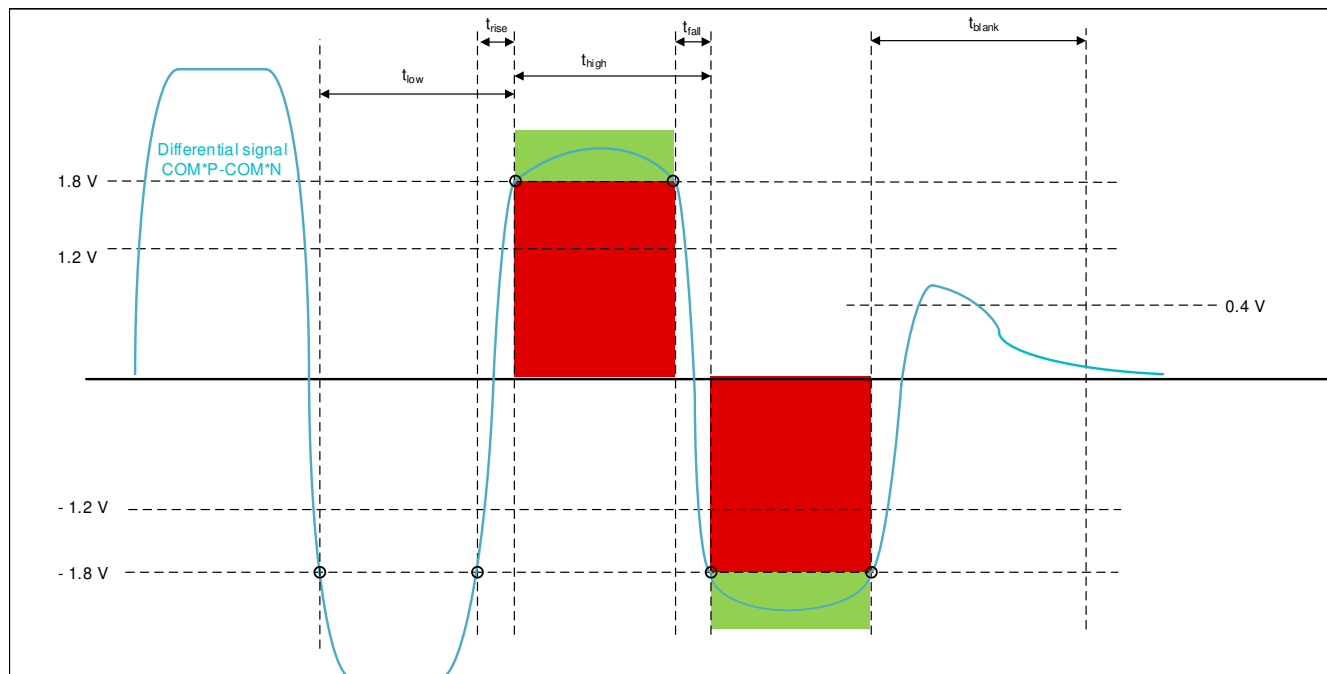


Figure 3-3. Differential Signal Eye Diagram - BQ79600

4 Debug Fault Registers

The bq7961X devices come equipped with many fault registers that help the user determine the cause and effect of signal noise between devices. Most importantly, we recommend always checking the `FAULT_SUMMARY` register to determine if an error occurs during communication. If the `FAULT_SUMMARY` flags a `FAULT_COMM` error then the next steps would be to read back the 3 `FAULT_COMM1/2/3` registers and `DEBUG` registers to identify the exact issue. Refer to the bq79616-Q1 data sheet section 9.6.4.14 to learn more about these registers and indicators.

5 Isolation Types

With the bq7961X family of devices, we offer several options for designing an isolation interface between stacked devices in a complete system. These include: 1) Transformer/Galvanic/Inductive 2) Capacitive + Choke and 3) Capacitive Only. Note that individual component properties and characteristics can affect signal integrity, robustness, and EMC performance. In the below sections, we go into detail of each isolation type and recommended component choices.

5.1 Transformer Isolation

For longer daisy chain cable lengths or highly noisy applications, the system may require the use of transformer isolation to provide the most robust form of communications. This design choice allows the daisy chain communications to be galvanically isolated and withstand higher sources of noise compared to capacitively coupled communications. It is recommended to use a certain set of parameters to meet automotive qualified temperature, voltage, and reliability ratings. It is recommended to choose parts that remain in the 150 – 600uH inductance range for best performance. Below is a list of recommended part numbers that were tested using the bq7961X devices. Choosing a part outside of this list or parameters would require thorough validation on the system designer side to ensure that the communication is robust over temperature, can withstand rapid communication transactions, and provides proper EMI filtering. Note that it is important and highly recommended to use the same transformer part number on both sides of a daisy chain interface (COML of top device and COMH of connected lower device).

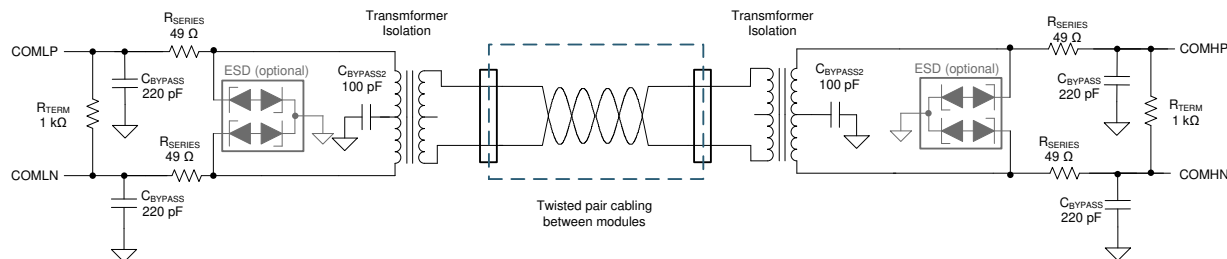


Figure 5-1. Transformer Isolation Across PCBs

Table 5-1. Recommended Transformer Part Numbers

Part Number	Supplier	Inductance (OCL) uH	Height (mm)
HMU1228NL	Pulse	150-370	6.85
HM2147NL	Pulse	580-1285	3.45
SM91502AL	Bourns	150-450	5.72

5.2 Capacitor Only Isolation

For low noise environments or devices mounted on the same pcb, typically it is best to choose capacitors only for isolation. We recommend these capacitors to be 2.2nF, high voltage rated (twice the voltage of the system's total pack voltage), and automotive qualified. Our reference schematic uses the 1206J2K00222KXR supplied by Knowles. It is possible to use a capacitor of up to 5.6nF for this isolation.

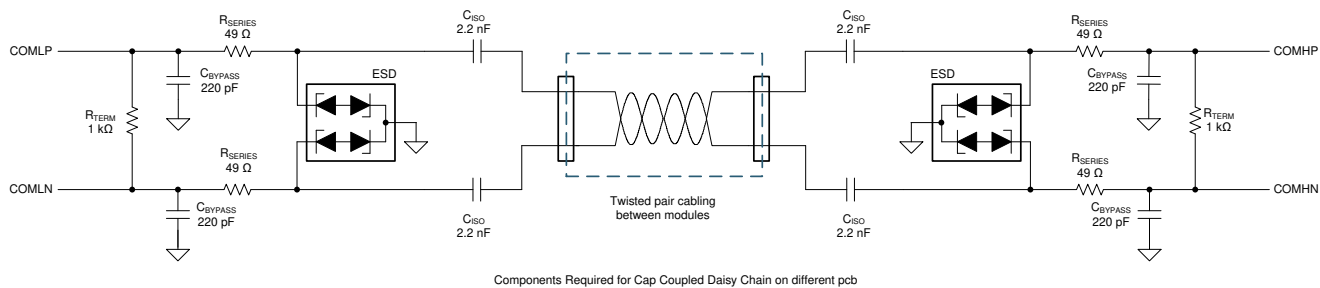
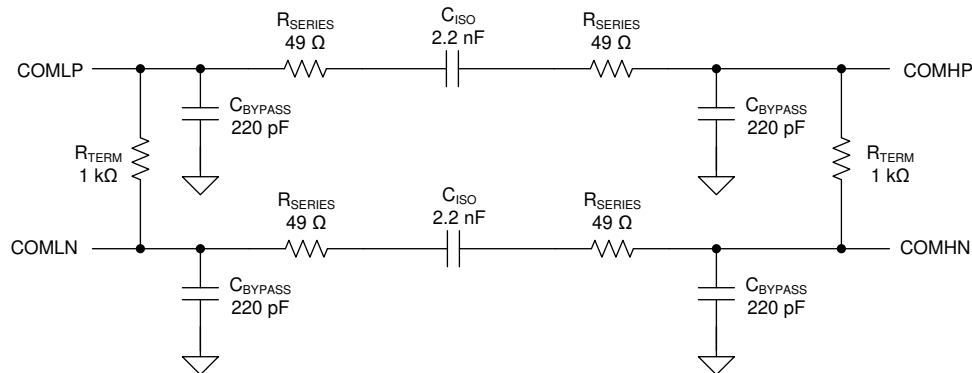


Figure 5-2. Capacitors Only Across PCBs



Components Required for Cap Coupled Daisy Chain on the same PCB

Figure 5-3. Capacitor Only on Same PCB

5.3 Capacitor + Choke Isolation

If the system uses capacitive based isolation, then it may be required for longer daisy chain cable lengths or highly noisy applications to use an additional common-mode choke filter in series with the capacitors. For these applications, it is recommended use an automotive grade 100 μH common-mode filter minimum for proper operation. To achieve the best performance in noisy environments, use dual common-mode filters (100 μH and 470 μH). For single common-mode filter applications, the TDK 51 μH 2.8k Ω choke (ACT45B-510-2P-TL003) is recommended. For dual common-mode filter applications, the TDK 100 μH 5.8 k Ω choke (ACT45B-101-2P-TL003) and Wurth 470 μH 2.2 k Ω (744242471) are recommended. Note that the common mode impedance specification is most critical to suppressing the higher noise sources. The same principle can be applied when determining choice of a ferrite bead.

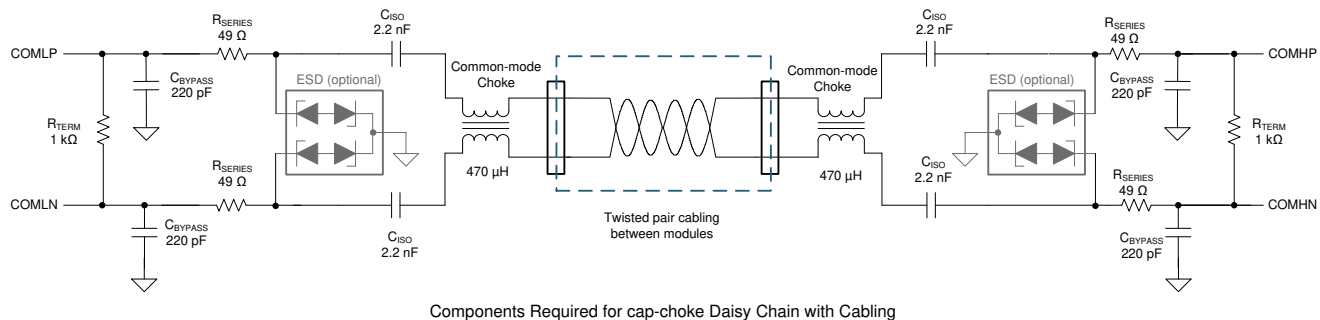


Figure 5-4. Capacitor + Choke Across PCBs

6 Mixed Isolation Circuits using bq79600-Q1

When using the bq79600-Q1 device as the communication bridge in combination with any variant of the bq79616-Q1 device as the stack device, it is important to validate the robustness of the isolation interface between the base and first stack device. This is because the devices are designed to support a mixed isolation type on the COMH side of the base vs. the COML of the stack device. Typically this means that the bq79600-Q1 will have a transformer from the above list, and then the user may select any of the 3 supported isolation types for the bq79616-Q1 stack device side (that is the same transformer part number on both sides if transformer isolation is desired). Every scenario with different transformer, cap, and choke part numbers mixed has not been tested so it is important for the user to ensure that the signal integrity is maintained after choosing their isolation components. Schematically, the key difference is that the decoupling capacitors on the bq79600-Q1 circuit are 100pF compared to 220pF for the bq79616-Q1 device.

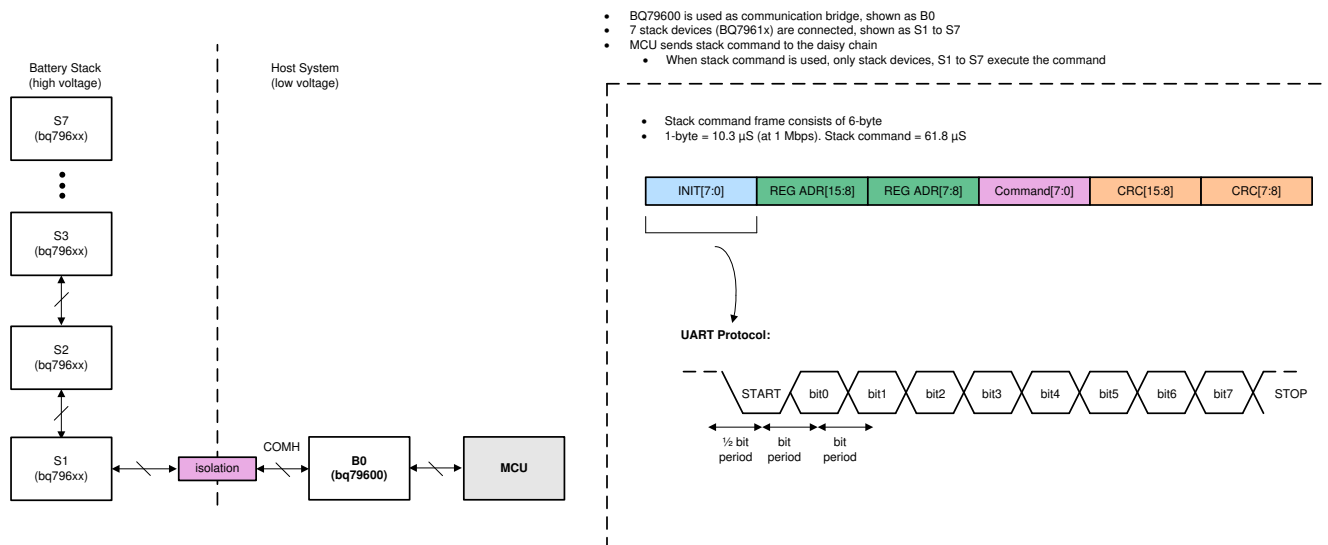


Figure 6-1. Example: BQ79600-Q1 Connected to BQ79616-Q1 Timing

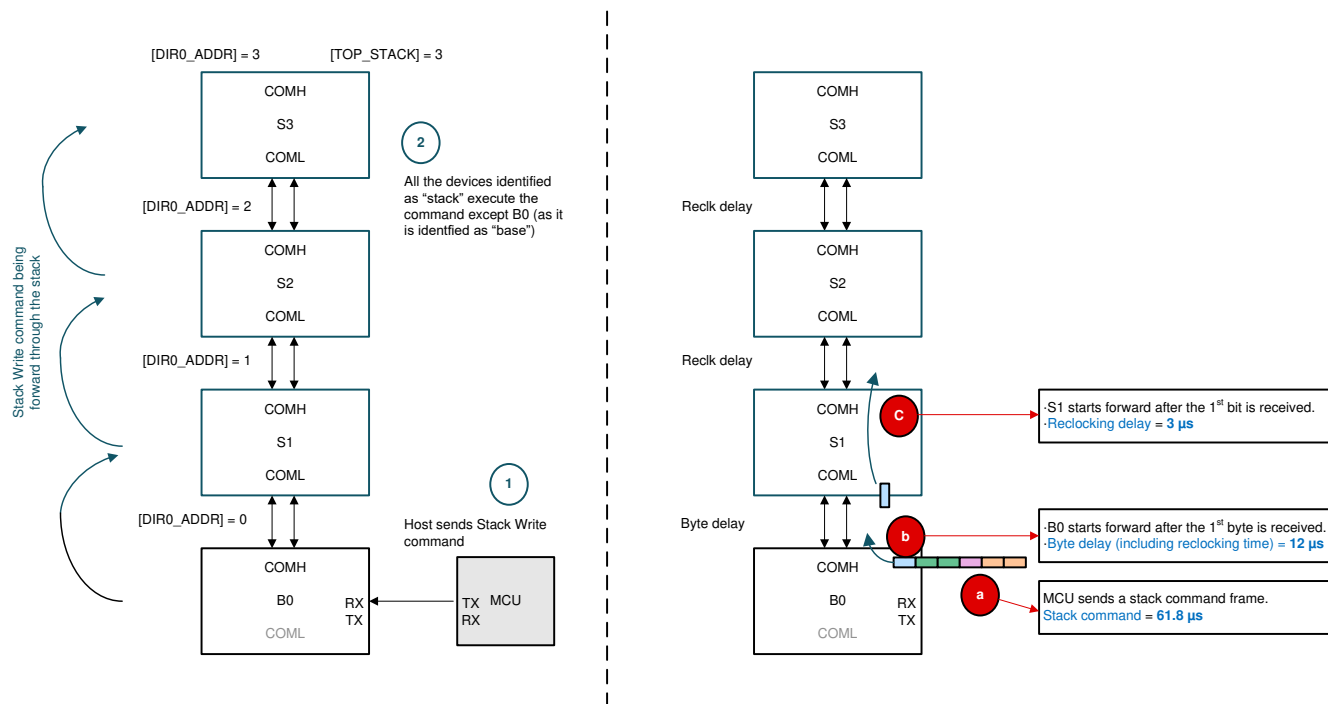


Figure 6-2. How a Stack Command is Sent

This is the tRECLK_DC spec in DS:
3µs typ, 5 µs max (will be using 3 µs for all calculations in this ppt)

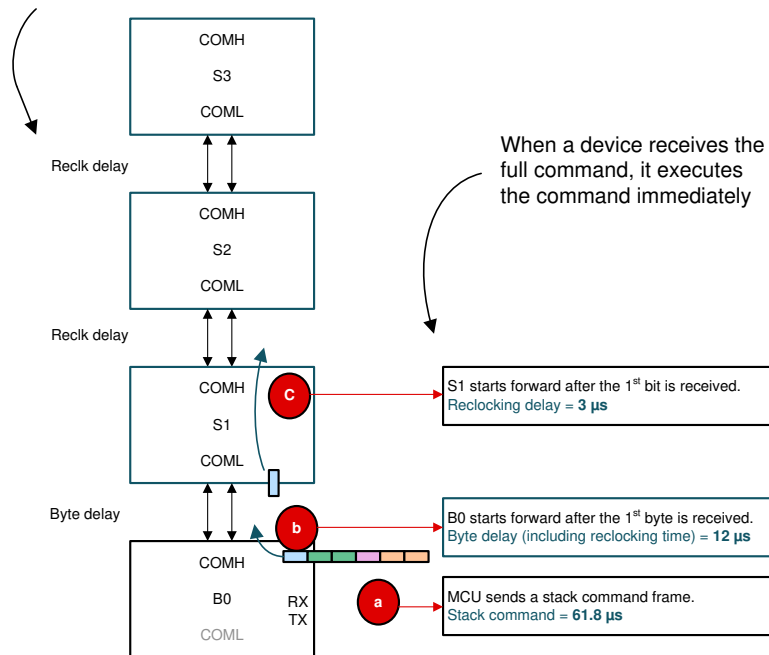
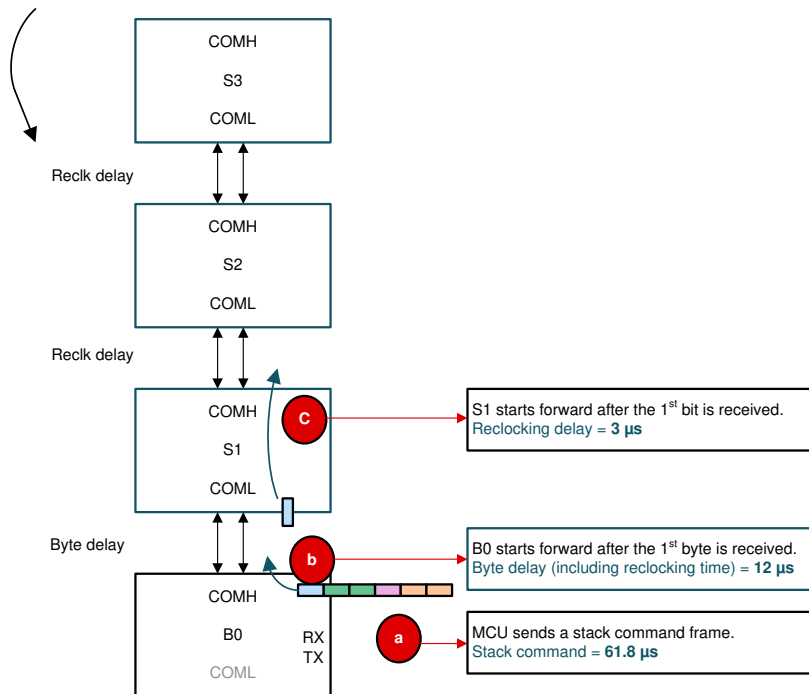


Figure 6-3. When Does Each Device Receive the Command?

This is the tRECLK_DC spec in DS:
3µs typ, 5µs max



	Time for each device to fully receive the stack command (count from MCU finishes sending out the command)
B0	Assume t0 = MCU finishes sending the last byte of the command = t0 = 0 µs
S1	Byte delay = 12 µs
S2	Byte delay + reclk = Byte delay + reclk * (N-1), where N = stack device # = 12 µs + 3 µs * (2-1) = 15 µs
S3	Byte delay + reclk * (N-1) = 12 µs + 3 µs * (3-1) = 18 µs

In this example, S3 is the last device to receive the command. Hence, once MCU finishes sending the command, all stack devices will receive the command after 19µs

Note: The command time (61.8 µs) shall be added to the timing above if counting from the start of MCU command

Figure 6-4. ??

7 Ring Architecture

The bq7961X daisy chain interface offers the option to use ring architecture. In this architecture, a cable break between two devices does not prevent communication to all upstream devices as in a normal non-ring scheme. When the host detects a broken communication, the device allows the host to switch the communication direction to communicate with devices on both sides of the break. This allows for safe operation until the break in the lines is repaired. The CONTROL1[DIR_SEL] controls the communication direction. The devices will reconfigure the COMH and COML ports depending on the [DIR_SEL] and the [TOP_STACK] settings. Auto-addressing procedure is needed to re-address the device address for the reverse communication direction. Example to change the communication direction to [DIR_SEL] = 1 to the entire daisy chain:

1. Host sends Single Device Write to change the base device [DIR_SEL] = 1. The base device will disable its COMH and enable its COML.
2. Host sends Broadcast Write Reverse Direction to clear the COMM_CTRL register settings on all devices.
3. Host sends Broadcast Write Reverse Direction to change the rest of the devices' [DIR_SEL] = 1. In this step, the entire daisy chain set up to transmitting communication in the [DIR_SEL] = 1 direction (that is, each device set up to transmit command frames sent by host from its COMH to its COML).
4. Host performs auto-addressing procedure to set up device address in the DIR1_ADDR register. Unless the devices have been reset, host can skip the dummy read/write steps to synchronize the DLL in the autoaddressing procedure.
5. Host sets up the new Top of Stack device and the new ToS device will disable its COML transmitter.

Once the device address in both communication directions is set up, the host can skip auto-address step when switching communication direction.

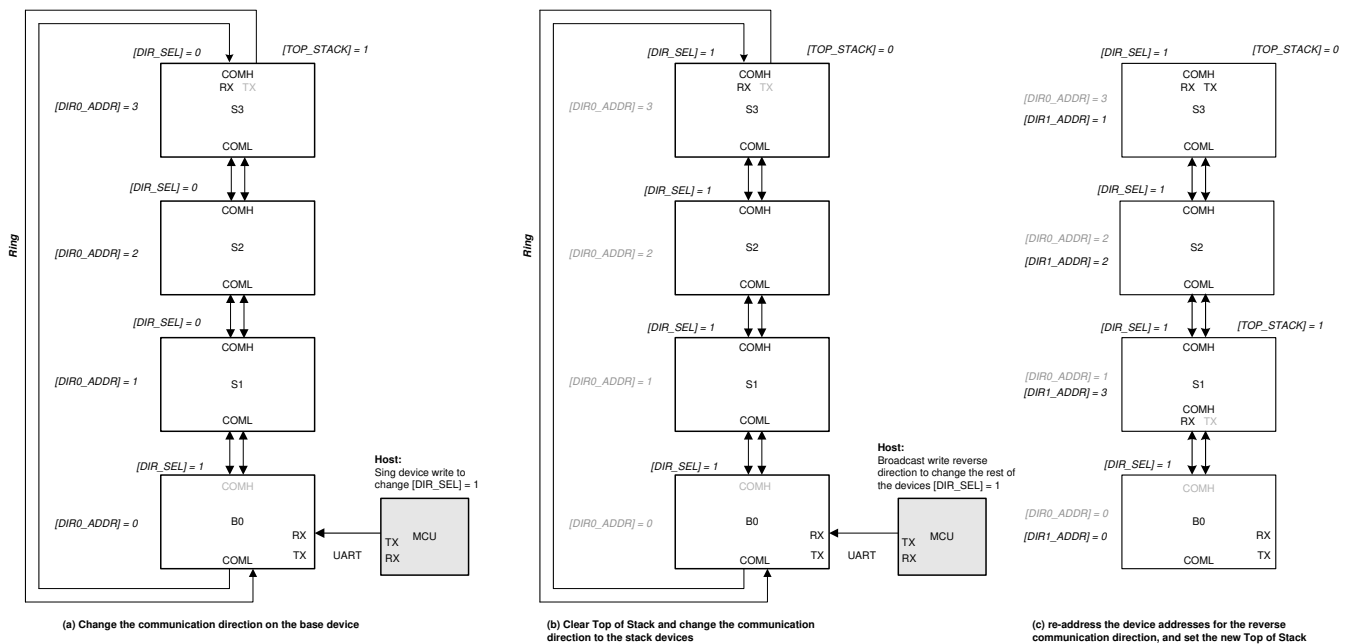
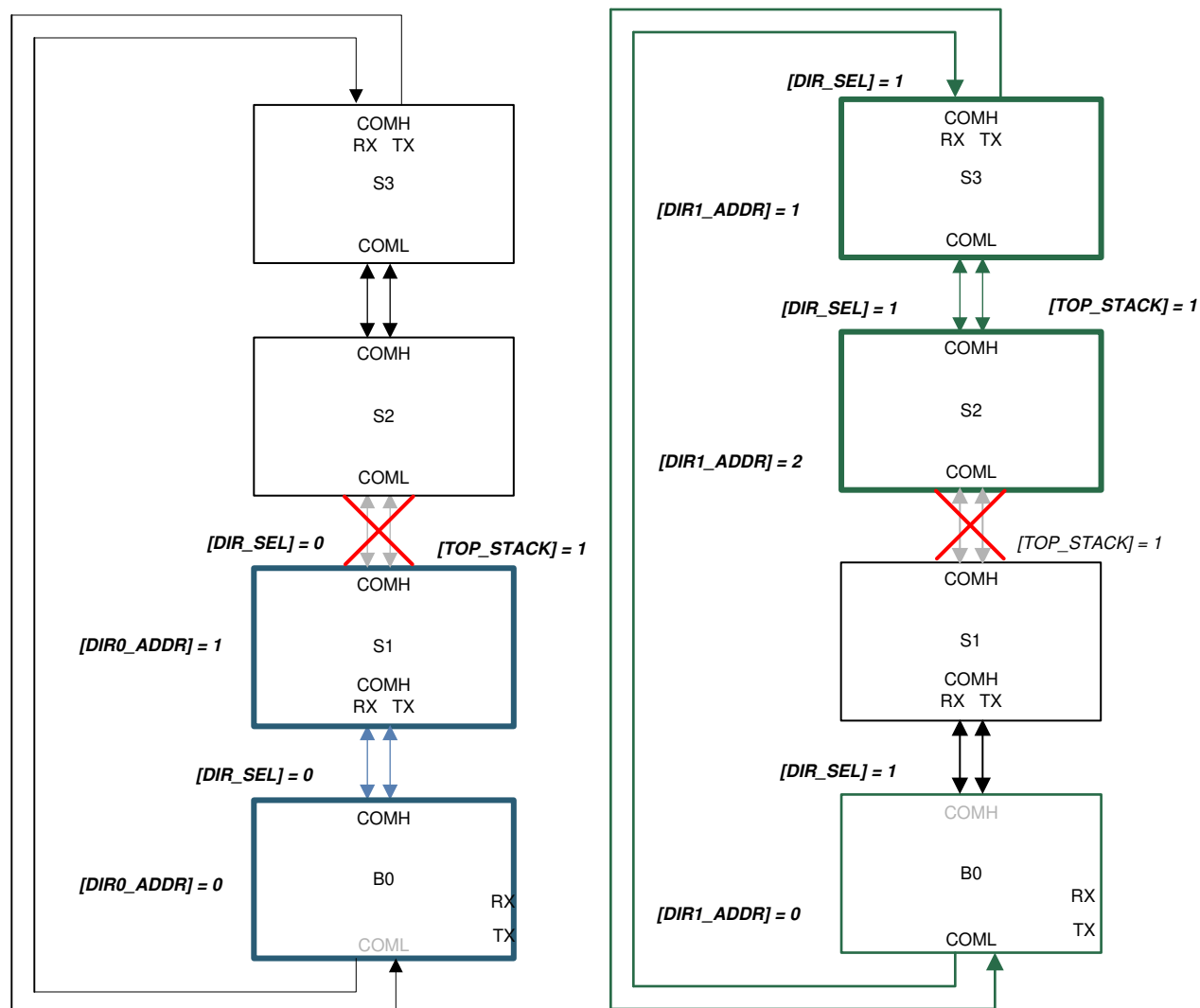


Figure 7-1. Example to Change Communication Direction in Daisy Chain

In a broken cable case, the host follows the same procedure to change the communication direction. To access all devices in the daisy chain, the host will have to communicate with [DIR_SEL] = 0 on some devices and communicate with [DIR_SEL] = 1 on other devices in the daisy chain. The chain will also have two ToS devices, one for each communication direction.



(a) Use [DIR_SEL] = 0 direction to communicate to S1

(a) Use [DIR_SEL] = 1 direction to communicate to S3 and S2

Figure 7-2. Using Ring Architecture to Access All Devices in a Broken Cable Case

8 Noise Immunity and Emissions

To determine that the bq7961X daisy chain architecture can be considered in automotive/EV applications, we perform several EMI/EMC tests to showcase the robustness. However, these results will vary based on schematical/pcb layout design choices, external noise sources, distance from UART/MCU connections, and calibration equipment. Reach out to TI sales team or support team representative to receive more details on these specifications passed. Below is a list of common practices in order to replicate best case results.:

1. Layout Recommendations in the bq79616-Q1 Data Sheet and Design Recommendations Application Report
2. Closely follow our reference EVM Schematic and Layout detailed in the EVM User's Guide
3. Common practices include isolating UART/MCU connections with ferrite, beads, filtering components.

9 Daisy Chain Cable Selection

Typically daisy chain testing is done with minimally short, symmetrical, matching impedance on-board pcb traces between devices or a 1 - 1.7m long twisted pair cabling between two separate pcbs or evaluation modules. However, depending on proper layout, isolation, and delay between commands - the communication protocol can support cabling up to 20-40m if needed but this would not be recommended for most robust solutions. The cabling material, twisting, and connectors can play a role in communication robustness but we do not provide strict requirements for these factors due to the flexibility provided to many different system architectures. However, testing has been completed with basic copper twisted pair wiring (no shielding) for many of our temperature and EMC testing. Although it may be best to use PP material rather than PVC material which changes the dielectric constant of the cable over time and temperature and thus has an impact on the transferred signals as opposed to more expensive, but better PP material. Typically a rule of thumb is to support at least 1-2 twists per cable inch and start the twisting very close to the connectors. This will help reduce external effects of EMI interference.

10 References

- bq79616-Q1 Data sheet (SLUSB75)
- bq79600-Q1 Data sheet (SLUSDS1)
- bq79616-Q1 Evaluation Module (SLUUC37)
- bq79616-Q1 Design Recommendations (SLVAE87)

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